

High-resolution kinetics of herbivore-induced plant volatile transfer reveal clocked response patterns in neighboring plants

Reviewed Preprint

Revised by authors after peer review.

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Reviewed preprint version 2

January 19, 2024 (this version)

Reviewed preprint version 1

August 25, 2023

Sent for peer review

June 27, 2023

Posted to preprint server

June 2, 2023

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Abstract

Volatiles emitted by herbivore-attacked plants (senders) can enhance defenses in neighboring plants (receivers), with important consequences for community dynamics. However, the temporal dynamics of this phenomenon remain poorly studied. Using a custom-built, high-throughput proton transfer reaction time-of-flight mass spectrometry (PTR-ToF-MS) system, we explored temporal patterns of volatile transfer and responses between herbivore-attacked and undamaged maize plants. We found that continuous exposure to natural blends of herbivore-induced volatiles results in clocked temporal response patterns in neighboring plants, characterized by an induced terpene burst at the onset of the second day of exposure. This delayed burst is not explained by terpene accumulation during the night, but coincides with delayed jasmonate accumulation in receiver plants. The delayed burst occurs independent of day : night light transitions and cannot be fully explained by sender volatile dynamics. Instead, it is the result of a stress memory from volatile exposure during the first day and secondary exposure to bioactive volatiles on the second day. Our study reveals that prolonged exposure to natural blends of stress-induced volatiles results in a response that integrates priming and direct induction into a distinct and predictable temporal response pattern. This provides an answer to the long-standing question of whether stress volatiles predominantly induce or prime plant defenses in neighboring plants, by revealing that they can do both in sequence.

eLife assessment

This **fundamental** study examines the effects of herbivory-induced maize volatiles on neighbouring plants and their responses over time. Measurements of volatile compound classes and gene expression in receiver plants exposed to these volatiles led to the conclusion that the delayed emission of certain terpenes in receiver plants after the onset of light may be a result of stress memory, highlighting the role of priming and induction in plant defences triggered by herbivore-induced plant volatiles. The evidence supporting the conclusions is **compelling**, with rigorous chemical assays of and state-of-the-art high throughput real time mass spectrometry. The work will be of broad interest to plant biologists and chemical ecologists.

Introduction

Plants rely on chemical cues to identify and mount appropriate responses to herbivores [1,2]. Among these cues are herbivore-induced plant volatiles (HIPVs) [3,4]. When perceived by undamaged plants, HIPVs can enhance plant defenses, and thus increase plant resistance to future herbivory [5–9]. HIPVs can directly induce plant defenses [7,8,10,11]. HIPVs can also prime plant defenses. In this case, HIPV-exposed plants respond more strongly and/or more rapidly to a secondary stimulus such as herbivore attack [5,7,8,12].

Plant responses to environmental cues depend on temporal patterns. For example, artificial damage that mimics spatiotemporal patterns of chewing herbivores induces a similar response to actual herbivore attack, whereas damage that ignores these patterns often does not [13]. The emission of HIPVs also follows discrete temporal dynamics [5,14], which, again, is likely to influence defense responses in receiver plants. Green leaf volatiles (GLVs), including (Z)-3-hexenyl acetate, (Z)-3-hexenal and (Z)-3-hexen-1-ol are catabolic products of the lipoxygenase/hydroperoxide lyase pathway and are emitted within minutes following damage [15,16]. Indole and terpenes are emitted within hours of herbivore attack [2,5]. Indole can mediate interactions between plants as well as between plants and other organisms [17–19]. Terpenes are a very diverse group of volatiles (> 80,000 known structures) that are also important as both direct and indirect plant defenses [6,20–22]. Indole and terpenes take longer to produce in comparison to GLVs and are less transiently emitted [2,5,15]. In maize, GLVs have been implicated in direct induction [8,10], and both GLVs and indole are known to prime defenses, and are thus considered as ‘bioactive’ in information transfer between plants [5,7].

Although it is clear that plants can either be primed or directly induced following HIPV exposure, if and how these two phenomena integrate in the context of a natural HIPV exposure sequence is unclear. Studies have yet to monitor induction and priming dynamics in real time and during continuous exposure to HIPVs. It is possible that, through time, continuous HIPV exposure may result in a self-enforced positive feedback loop, whereby priming enhances induction and induction serves as a priming event for subsequent induction.

Further, temporal trends might be complicated by dynamic environmental conditions, such as light fluctuations; over a 24 hr period, plants can experience both full sun and darkness. In darkness, the emission of many volatiles, especially those released via stomata, is severely hindered [23–26].

In order to understand the kinetics of information transfer between plants and the importance of dynamic temporal emission patterns, we monitored the emission kinetics of terpenes as a marker of defense activation in maize plants that were exposed to HIPVs. We leveraged a highly temporally-resolved volatile multiplexing system that allowed us to track continuous volatile emissions in sender and receiver plants for the first time in real time. We further measured foliar terpene pools, stress hormones and defense gene expression in receiver plants, and determined the impact of day : night light transitions in the observed defense induction kinetics. Finally, we measured defenses upon sequences of short-term HIPV and GLV exposure to unravel the relative importance of direct induction and priming. Together, our experiments reveal how the natural kinetics of herbivore attack result in a clocked defense response in neighboring plants via volatile information transfer.

Results

Delayed burst in induced volatile emissions in plants exposed to volatiles of an herbivore-attacked neighbor

When maize plants are exposed to volatiles from herbivore-attacked neighbors, they start to release terpenes [7,11]. To understand the temporal dynamics of this phenomenon, we conducted a detailed time-course analysis of terpene release by receiver plants exposed to volatiles from sender plants under attack by *Spodoptera exigua* caterpillars (Fig 1). Herbivore feeding began around 12:00 and within 2 hr, we detected a small induction of sesquiterpenes [$C_{15}H_{21}^+$, $m/z = 205.20$], monoterpenes [$C_{10}H_{17}^+$, $m/z = 137.13$], 4,8-dimethylnona-1,3,7-triene (DMNT) [$C_{11}H_{19}^+$, $m/z = 151.15$] and 4,8,12-trimethyltrideca-1,3,7,11-tetraene (TMTT) [$C_{16}H_{27}^+$, $m/z = 219.21$] in undamaged receiver plants. Volatile emission was severely impaired during the night, regardless of treatment. Interestingly, as soon as the light was restored, we observed a strong burst of terpene release from receiver plants, which was 2-5 times higher than the release the day before (Fig 1). Thus, exposure to volatiles from an herbivore-attacked plant triggers a delayed burst in terpene emission in neighboring plants.

The delayed burst in terpene emission is not explained by overaccumulation during the night

Why do receiver plants show a delayed burst in volatile-induced terpenoid release? During the night, plants close their stomata to limit water loss [27], which may also impair terpene release, thus leading to an accumulation of terpenes in the leaves and a burst once stomata open again [23,28]. To determine if terpenes accumulate above daytime levels in maize leaves of HIPV-exposed plants during the night, we measured internal foliar pools of the major sesquiterpenes, β -caryophyllene, β -farnesene, and α -bergamotene, as well as the homoterpene, TMTT over time. Internal pools of β -caryophyllene, β -farnesene and TMTT were marginally induced after 3 hours, but only became significantly induced after 8 hr of exposure to HIPVs (Fig 2A, 2B and 2D, Table 1). For all three, accumulation remained higher during the night (16.75 hr) as well as the following day. α -bergamotene only began accumulating in leaves on the second day (Fig 2C, Table 1). Thus, internal terpene pools remain comparable between night and day, suggesting that terpenes do not continue accumulating during the night and that, even when terpenes are emitted in large amounts on day two, internal pools do not drop below nighttime levels and that volatiles. Nevertheless, the volatiles that are present during the night, may be released suddenly the second day.

Some control plants accumulated measurable amounts of β -caryophyllene at night, likely due to biological and experimental variation. This type of variation was not visible in total sesquiterpene emissions measured by PTR-MS (Fig 1), likely because β -caryophyllene is a minor sesquiterpene in B73 [29] and closed stomata block sesquiterpene emission at night [23]. To get insight into terpene biosynthesis we measured the expression of terpene synthases in receiver plants. In maize, farnesene pyrophosphate synthase 3 (FPPS3), terpene synthase 2 and 10 (TPS2 and TPS10) as well as dimethylnonatriene/trimethyltetradecatetraene synthase (CYP92C5) are rate limiting for terpene production [30–32]. All genes were induced by HIPV exposure during daytime, but not during the night. FPPS3 was slightly induced after 3 hr and significantly induced after 8 hr of HIPV exposure (Fig 2E, Table 1). TPS2 was slightly induced 3, 8 and 22 hr after the onset of HIPV exposure (Fig 2F, Table 1). TPS10 was significantly induced after 3 and 22 hr of HIPV exposure (Fig 2G, Table 1). CYP92C5 was only significantly induced after 3 hr of HIPV exposure (Fig 2H, Table 1). The patterns of volatile terpene biosynthesis do not support a scenario where the terpene burst on the second day is due to continued biosynthesis but lack of emission during the night.

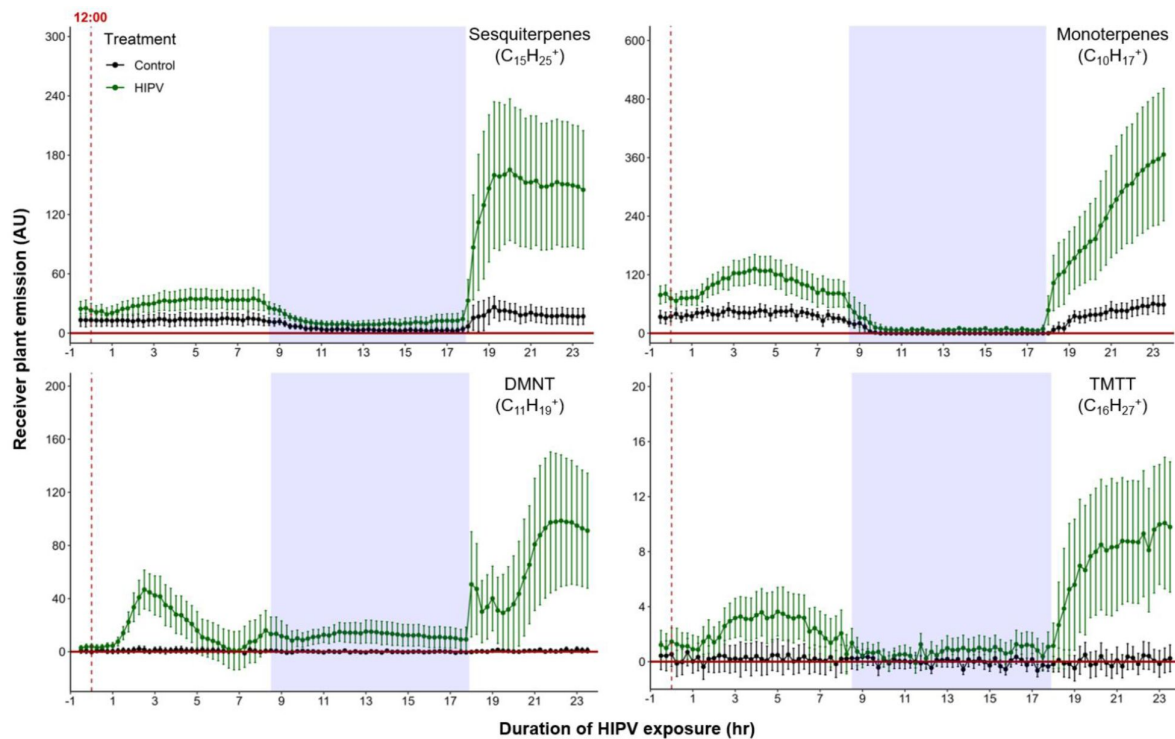


Figure 1.

Delayed burst in induced volatile emissions in plants exposed to volatiles of a herbivore-attacked neighbor. Emission kinetics of herbivore-induced plant volatile (HIPV)-induced terpenes in undamaged receiver plants are shown. Dark green points represent mean emission of herbivore-damaged sender plants connected to undamaged receiver plants, with the emissions from damaged sender plants only subtracted. Black points represent undamaged sender plants connected to undamaged receiver plants, with the emissions from undamaged sender plants only subtracted. Blue rectangles represent the dark phase. Abbreviations. DMNT, 4,8-dimethylnona-1,3,7-triene; TMTT, 4,8,12-trimethyltrideca-1,3,7,11-tetraene. Error bars = SE. $n = 8-10$. Compounds were identified based on their molecular weight + 1, as all compounds were protonated. Sesquiterpenes: $m/z = 205.20$, Monoterpenes: $m/z = 137.13$, DMNT: $m/z = 151.15$, TMTT: $m/z = 219.21$.

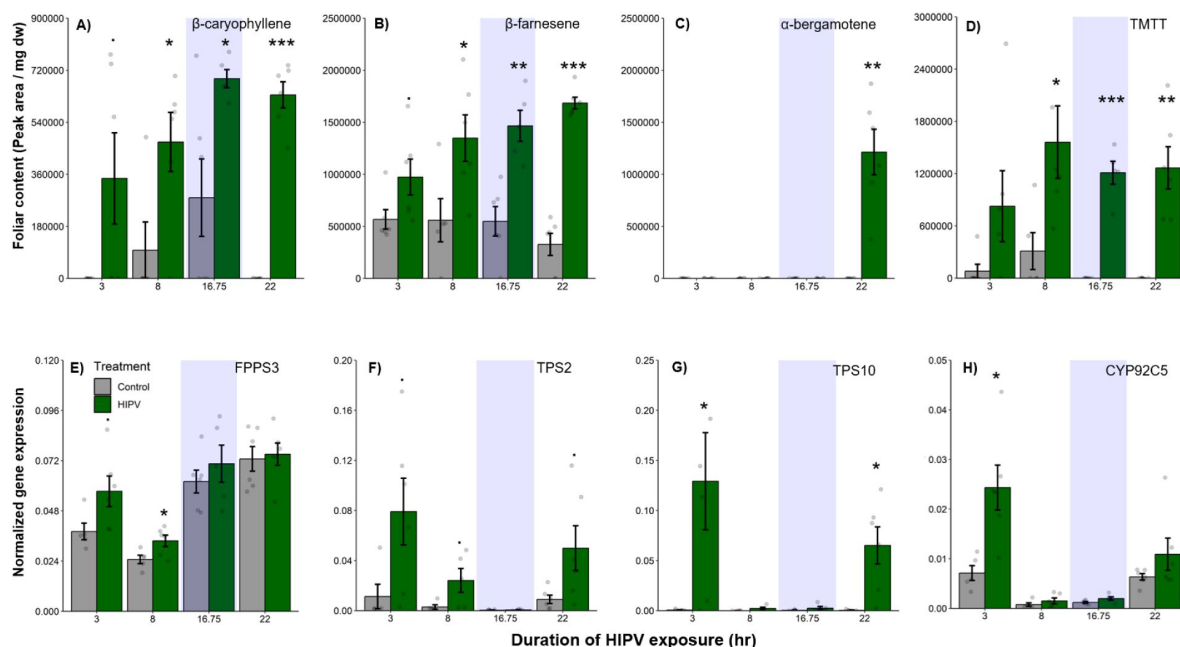


Figure 2.

The delayed burst in terpene emission is not explained by terpene overaccumulation during the night. Accumulation of terpenes and induction of terpene biosynthesis genes in receiver plants was measured over time. A-D) Internal foliar pools of terpenes in receiver plants. E-H) Expression of terpene biosynthesis genes in receiver plants. Dark green bars represent receiver plants exposed to herbivore-induced plant volatiles (HIPVs) and light grey bars represent receiver plants connected to undamaged sender plants. Blue rectangles represent the dark phase. Abbreviations: TMTT, 4,8,12-trimethyltrideca-1,3,7,11-tetraene; FPPS3, Farnesyl pyrophosphate synthase 3; TPS2, Terpene synthase 2; TPS10, Terpene synthase 10; CYP92C5, Dimethylnonatriene/trimethyltetradecatetraene synthase. . = $p < 0.1$, * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$ as determined by a Welch's two-sample t -test. Bars = mean $\pm$ SE. $n = 4-6$.

Response	Time											
	3 hr			8 hr			16.75 hr			22 hr		
	<i>t</i>	<i>p</i>	df	<i>t</i>	<i>p</i>	df	<i>t</i>	<i>p</i>	df	<i>t</i>	<i>p</i>	df
β-car	-2.20	<u>0.08</u>	5	-2.65	0.03	9	-3.00	0.03	6	-14.05	<0.001	5
β-farn	-2.08	<u>0.07</u>	8	-2.60	0.03	9	-4.47	0.002	9	-11.38	<0.001	8
α-berg	-	-	-	-	-	-	-	-	-	-5.56	0.003	5
TMTT	-1.80	0.13	5	-2.68	0.03	7	-9.18	<0.001	4	-5.23	0.003	5
FPPS3	-2.31	<u>0.05</u>	8	-2.62	0.03	9	-0.82	0.44	7	-0.28	0.78	10
TPS2	-2.39	<u>0.05</u>	6	-2.19	<u>0.09</u>	4	-0.62	0.55	8	-2.24	<u>0.07</u>	5
TPS10	-2.65	0.05	5	-1.97	0.12	4	-1.34	0.25	4	-3.50	0.02	5
CYP92C5	-3.62	0.01	6	-1.11	0.30	8	-1.94	0.11	5	-1.38	0.22	5
OPDA	-0.26	0.80	4	0.07	0.95	5	0.65	0.55	4	-4.18	0.008	5
JA	0.32	0.78	2	1.22	0.31	3	0.91	0.41	4	-1.40	0.22	5
JA-Ile	-0.48	0.68	2	1.21	0.31	3	0.24	0.82	5	-2.98	0.04	4
OPR7	-5.70	0.002	5	-2.36	<u>0.05</u>	7	1.32	0.22	10	-3.01	0.02	7

Table 1.

Welch's two-sample *t*-test results comparing foliar terpene pools, biosynthesis genes and phytohormone levels between HIPV-exposed and control receiver plants. Bold values: $p < 0.05$, underlined values: $p < 0.1$. Abbreviations: β-car, β-caryophyllene; β-farn, β-farnesene; α-berg, α-bergamotene; TMTT, 4,8,12- trimethyltrideca-1,3,7,11-tetraene; FPPS3, Farnesyl pyrophosphate synthase 3; TPS2, Terpene synthase 2; TPS10, Terpene synthase 10; CYP92C5, Dimethylnonatriene/trimethyltetradecatetraene synthase; OPDA = 12- oxophytodienoic acid; JA = jasmonic acid; JA-Ile = jasmonic acid-isoleucine; OPR7 = oxo-phytodienoate reductase 7.

The delayed burst in terpenoid emission is associated with clocked jasmonate production

Volatile release in maize is regulated by jasmonates [33,34]. To understand whether the delayed terpene burst is associated with jasmonate signaling, we measured the levels of 12-oxo-phytodienoic acid (OPDA), jasmonic acid (JA) and jasmonic acid-isoleucine (JA-Ile), as well as the expression of oxo-phytodienoate reductase 7 (OPR7), which is critical for JA biosynthesis [35]. We found a significant induction of OPDA, JA-Ile production and OPR7 expression at the beginning of the second day, 22 hr after the onset of volatile exposure (Fig 3, Table 1). Thus, jasmonate production is temporally aligned with the delayed terpene burst at the onset of the second day.

The delayed volatile burst is conserved under continuous light

To test whether the delayed burst in terpene volatiles is linked to light : dark transitions, we exposed maize plants to HIPVs under continuous light. Similar to exposure during a normal light regime, the largest burst of terpene emission occurred ca. 12-18 hr after the onset of volatile exposure (Fig 4), suggesting that the temporal delay in volatile emission in receiver plants is not dependent on light fluctuations, but is otherwise clocked (Fig 4).

In order to control for potential differences in induction capacity at different times of day, we designed a second continuous light experiment. However, instead of starting the herbivory treatment at ca. 12:00 hr, we started it around 20:00 hr. Similarly, we observed an initial bump in terpene emission followed by a larger burst 10-12 hr after HIPV exposure (Supplemental Fig 1). In both continuous light experiments the 'second day' burst occurred somewhat earlier than under normal light conditions (Fig 4 and Fig 1). It is possible that stomata, which are closed in the dark, delay the onset of the terpene burst under dark : light transitions.

The delayed volatile burst cannot be fully explained by the emission kinetics of bioactive herbivory-induced volatiles

A simple explanation for the delayed volatile burst may be that the bioactive volatiles are more strongly emitted from sender plants at the onset of day two, thus triggering a stronger volatile response in the receiver plants at this time. To test this hypothesis, we analyzed volatile emission kinetics of *S. exigua*-infested plants over time. We focused our analysis on GLVs and indole, which are known to induce and/or prime volatile release in neighboring maize plants [7,8,10,11]. Terpenes are not known to prime or induce volatile release in maize, and were thus not included in the sender plant analysis [11]. We analyzed GLV and indole emission data of the sender plants from the experiments depicted in Fig 1 and Fig 4 (Fig 5), and then correlated their emission with the induction of terpenes in receiver plants (Fig. 6). We found strong positive correlations between hexenyl acetate, hexenal, hexen-1-ol and indole emissions in sender plants and terpene responses in receiver plants when plants had a dark period (Fig 6). Interestingly, this was not the case under continuous light, where the apex of bioactive volatile emission in sender plants was not temporally aligned with the apex of terpene responses in receiver plants (Fig 6). Sender plant emissions even showed slight negative correlations with terpene induction in receiver plants under continuous light. Thus, the emission kinetics of bioactive volatiles from sender plants cannot fully explain the delayed terpene burst.

The combination of direct induction and priming can explain the delayed terpene burst in receiver plants

Based on the above observations, we reasoned that, under natural conditions, the strong delayed volatile burst may be due to the interaction of an initial volatile burst that primes plants for higher volatile release at the onset of the second day, and a secondary trigger in the form of bioactive

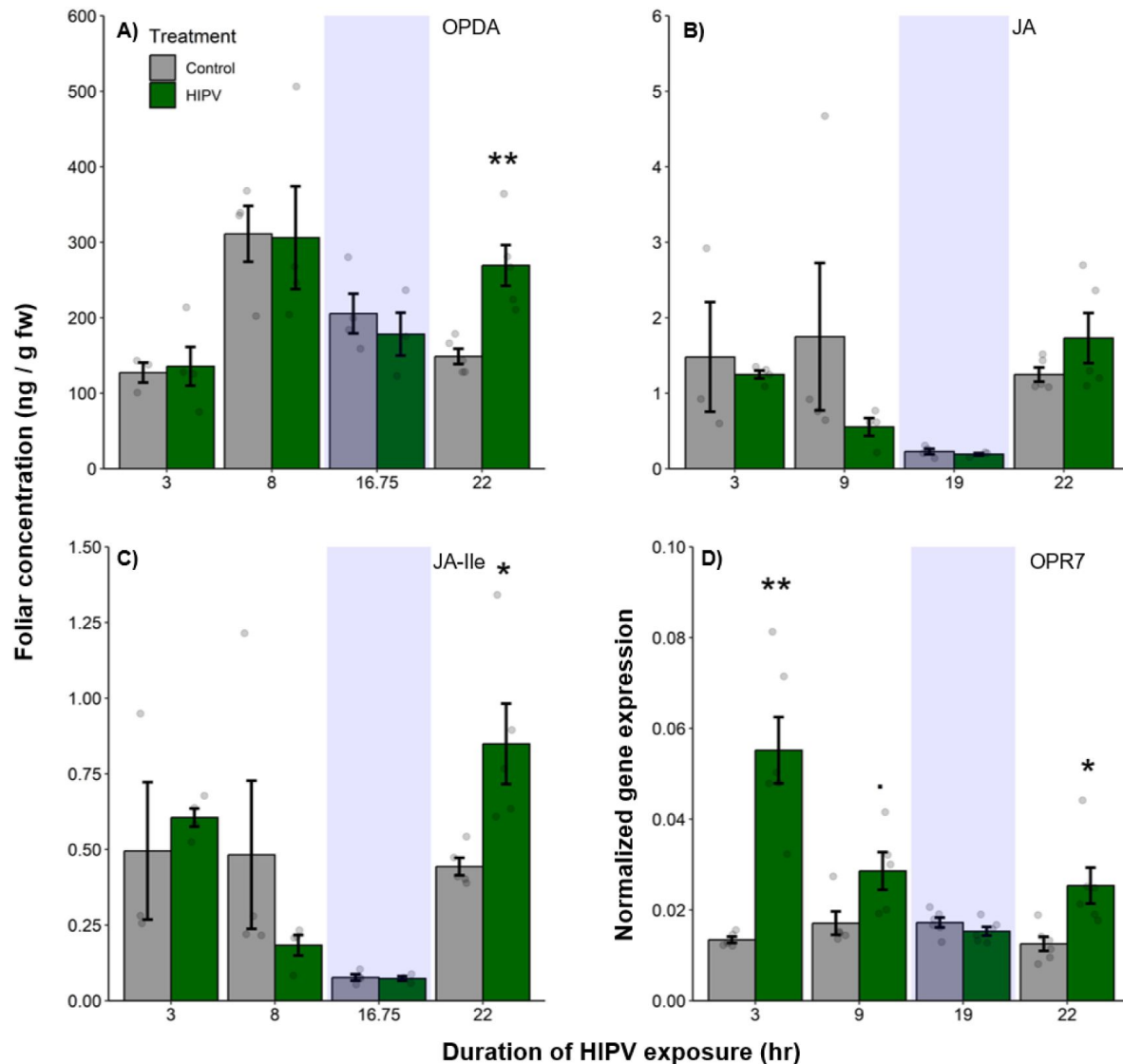


Figure 3.

The delayed burst in terpene emission is associated with clocked jasmonate production. Foliar jasmonate concentrations (A-C) and jasmonate biosynthesis (OPR7; D) in receiver plants over time are shown. Dark green bars represent receiver plants exposed to herbivore-induced plant volatiles (HIPVs) and light grey bars represent receiver plants connected to undamaged sender plants. Blue rectangles represent the dark phase. Abbreviations. OPDA = 12-oxophytodienoic acid, JA = jasmonic acid, JA-Ile = jasmonic acid-isoleucine, OPR7 = oxo-phytodieneoate reductase 7. . = $p < 0.1$, * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$ as determined by a Welch's two-sample t -test. Bars = mean $\pm$ SE. $n = 3-6$.

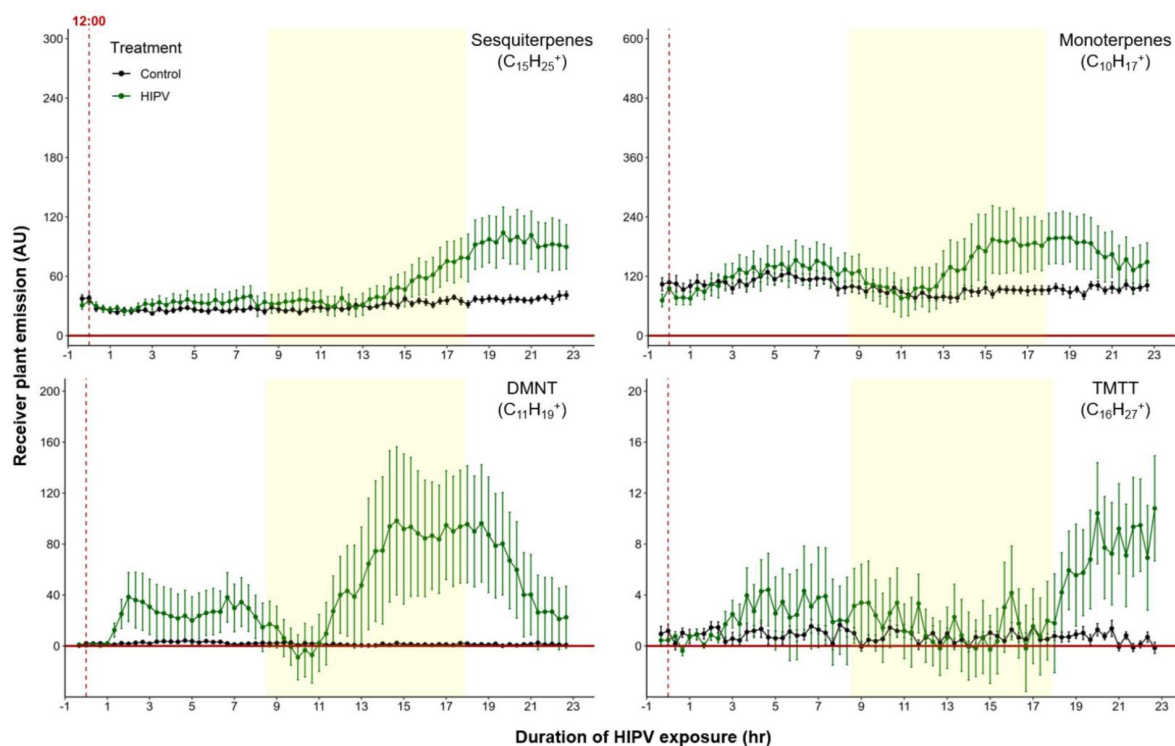


Figure 4.

The delayed volatile burst is conserved under continuous light. Emission kinetics of herbivore-induced plant volatiles (HIPV)-induced terpenes in undamaged receiver plants under continuous light are shown. Plants were grown under normal light conditions. Lights were left on continuously following the start of the treatment. Dark green points represent the mean emission of herbivore damaged sender plants connected to undamaged receiver plants with the emissions from damaged sender plants only subtracted. Black points represent undamaged sender plants connected to undamaged receiver plants, with the emissions from undamaged sender plants only subtracted. Yellow rectangles represent when the lights would typically be turned off. Abbreviations: DMNT, 4,8- dimethylnona-1,3,7-triene; TMTT, 4,8,12-trimethyltrideca-1,3,7,11- tetraene. Error bars = SE. $n = 8-10$. Compounds were identified based on their molecular weight + 1, as all compounds were protonated. Sesquiterpenes: $m/z = 205.20$, Monoterpenes: $m/z = 137.13$, DMNT: $m/z = 151.15$, TMTT: $m/z = 219.21$.

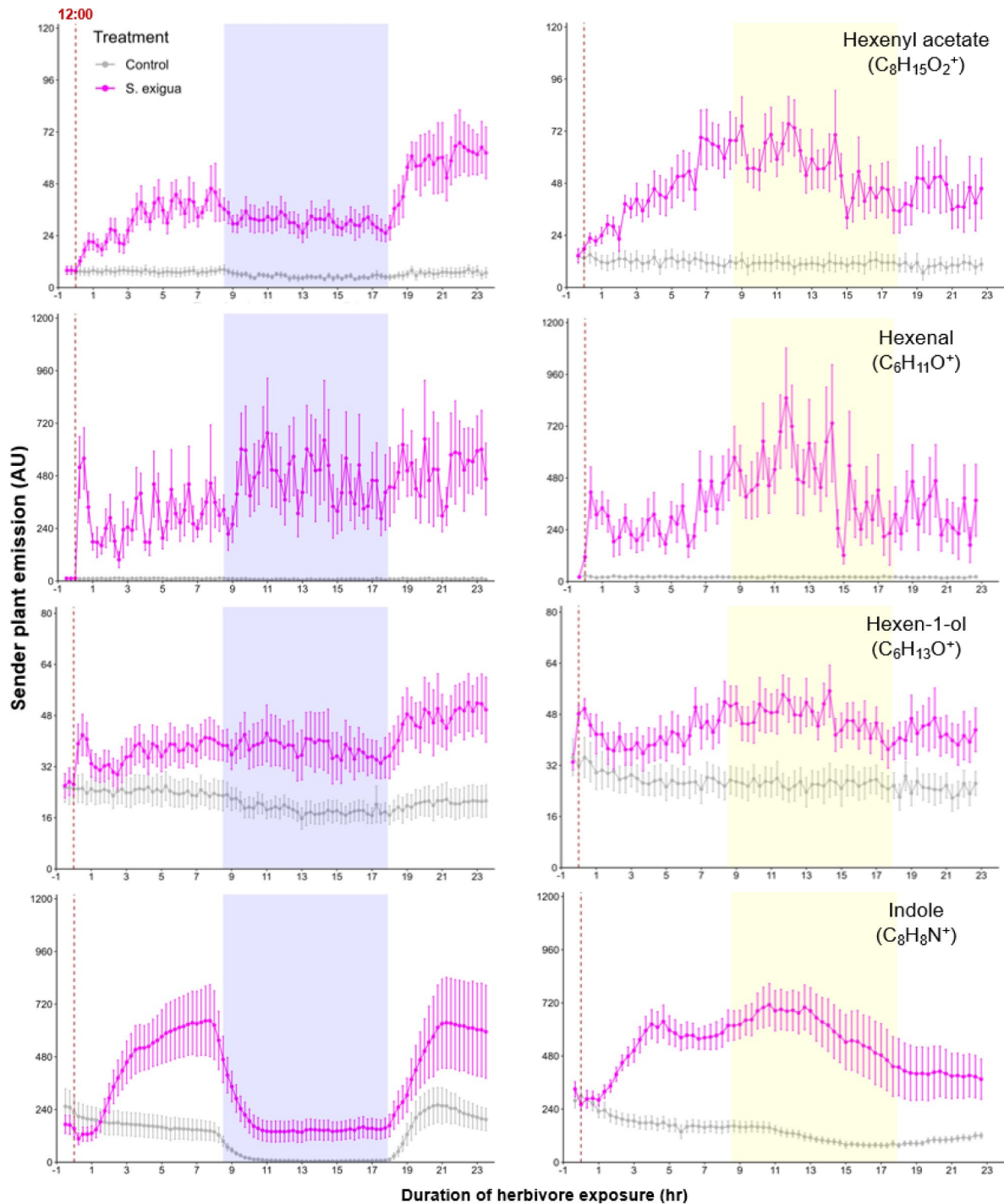


Figure 5.

Green leaf volatile (GLV) and indole emissions in *Spodoptera exigua*-damaged plants. Light green points represent the mean emission of herbivore-damaged sender plants. Grey points represent mean emissions of undamaged sender plants. Blue rectangles represent the dark phase. Yellow rectangles represent when the lights would typically be turned off. Continuous light-exposed plants were grown under normal light conditions, however lights were left on continuously following the start of the treatment. Error bars = SE. $n = 8-10$. Compounds were identified based on their molecular weight + 1, as all compounds were protonated. Sesquiterpenes: $m/z = 205.20$, Monoterpenes: $m/z = 137.13$, DMNT: $m/z = 151.15$, TMTT: $m/z = 219.21$.

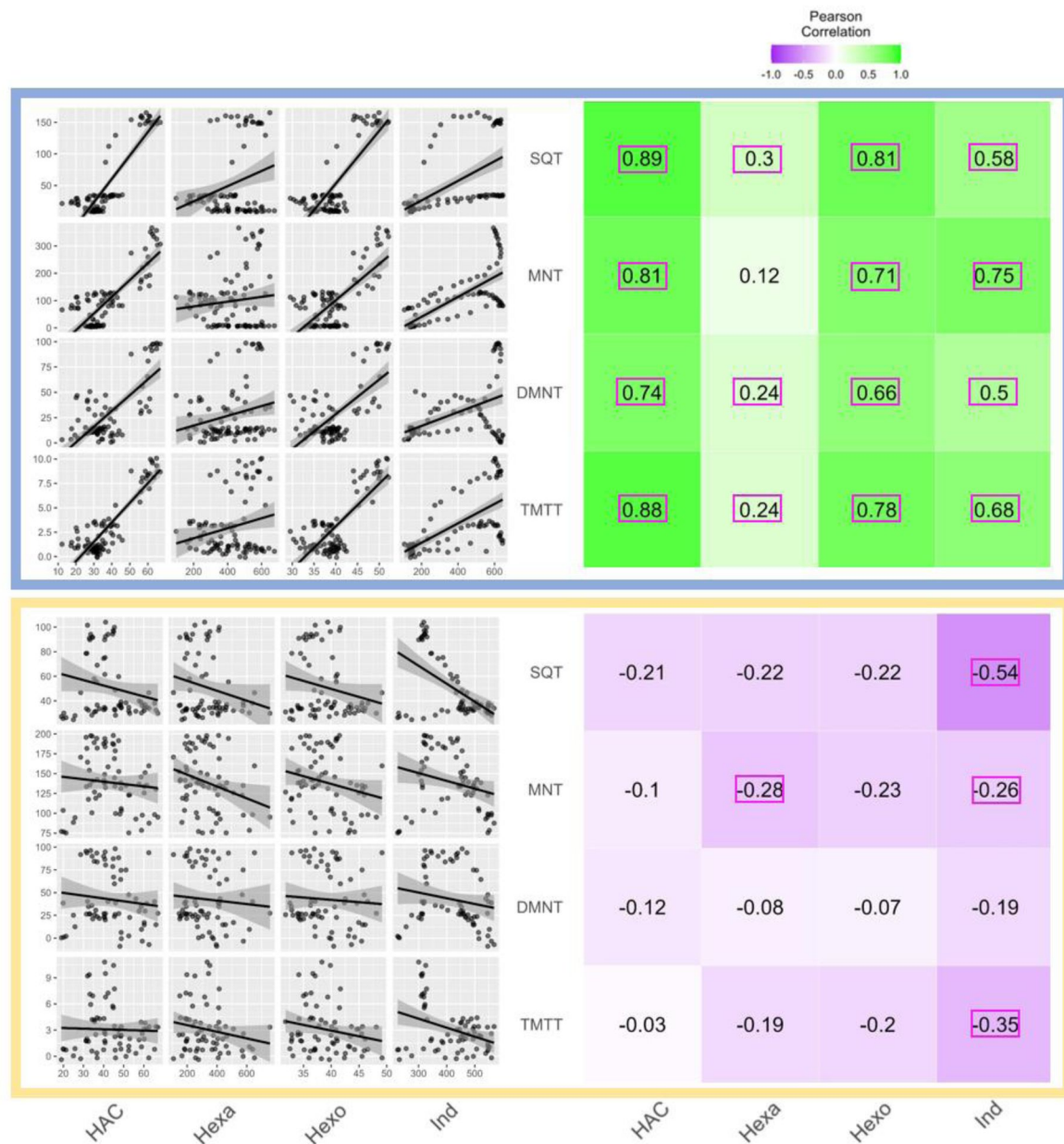


Figure 6.

Receiver plant terpene emissions are tightly correlated with bioactive sender plant signals under normal light conditions. The left panels depict scatter plot correlation matrices of bioactive volatile emissions from damaged sender plants and terpene emissions from (HIPV)-exposed receiver plants. Only data from the first measurement following the addition of herbivores to sender plants are included. Upper scatter plot (in blue box) shows correlations under normal light conditions and lower scatter plot (yellow box) depicts correlations under continuous light. Continuous light-exposed plants were grown under normal light conditions, however, lights were left on continuously following the start of the treatment. Each black point represents the mean value of all individuals at a given time point after herbivory began. Panels on the right hand side depict heat maps based on the value of Pearson's correlation coefficient between two given compounds. Numbers in the center of each square are Pearson's correlation coefficient. Correlation coefficients contained in a pink rectangles indicate a significant correlation ($p < 0.05$). Hexenyl acetate (HAC), Hexenal (Hexa), Hexen-1-ol (Hexo) and indole (Ind) were from sender plants, and sesquiterpenes (SQT), monoterpenes (MNT), 4,8-dimethylnona-1,3,7-triene (DMNT) and 4,8,12-trimethyltrideca-1,3,7,11-tetraene (TMTT) were from receiver plants.

volatiles released from sender plants at the onset of the second day. To test this hypothesis, we exposed receiver plants to volatiles from control plants or volatiles from herbivory-induced plants for 1.25 hr. We then disconnected the receiver plants and exposed them to clean air for 17 hr, until the beginning of the next day. Half of the plants were then exposed to (Z)-3-hexenyl acetate (HAC) as a secondary trigger. HAC was selected as an inducer as it showed the strongest correlation with all measured terpenes (Fig 6). Following the short exposure to HIPVs, we detected a generally small, but significant, induction of sesquiterpenes, monoterpenes, DMNT and TMTT emissions in receiver plants (Fig 7, Table 2). During the night, no differences between treatments were detected any more. At the beginning of the next day, we found a slight induction in terpene emissions, most apparent for sesquiterpenes and TMTT, in plants that had been exposed to herbivory induced volatiles 18 hr prior (Fig 7, Table 2). (Z)-3-hexenyl acetate exposure at the beginning of the next day also induced terpene emissions. Furthermore, (Z)-3-hexenyl acetate induced a stronger release of all terpenes in plants that had been exposed to HIPVs the previous day. In an independent experiment, we also tested whether HAC sensitivity *per se* may be higher at the start of the light period. We exposed maize seedlings to HAC dispensers either 30 min or 4 hr 30 min after lights came on. We found induction in terpene emission to be similar regardless of when induction began (Supplemental Fig 2). Thus, exposure to HIPVs prompts maize plants to release more terpenes the next day and also primes maize plants to respond more strongly to a secondary HIPV trigger. Together, these two phenomena result in a pronounced terpene burst.

Discussion

Herbivore-induced plant volatiles (HIPVs) play an important role in mediating interactions between damaged and undamaged plants [2]. However, the kinetics and temporal dynamics of this information transfer remain poorly understood. We show that, somewhat surprisingly, upon continuous exposure to HIPVs, receiver plants show a pronounced activation of defenses at the onset of the second day. We find that this is the result of volatile-mediated priming on the first day that yields a clocked response to the volatiles that are perceived on the next day. Here, we discuss the mechanisms and biological implications of this phenomenon.

Plants are well known to respond to herbivore-induced volatiles by increasing their own defenses. Defense activation can happen directly, with plants increasing their defense hormone production and volatile release [8,10], and/or via priming effects, with plants increasing their defenses more strongly upon a secondary stimulus [5,7]. Here, we show that these two phenomena operate together to trigger strong defense activation in receiver plants with predictable temporal kinetics. So far, studies in maize and other plants showing priming effects have often exposed receiver plants to HIPVs for several hours, overnight, or even for multiple days, and induced them, e.g. by simulated herbivory, on the following day in the absence of HIPVs [5,8,36–38]. Indeed, such a setup would reveal clear priming, with minor or no direct induction by the volatile treatment. Our findings demonstrate that more natural continuous exposure to volatiles results in the same pattern without the need for another stimulus. We thus conclude that HIPVs are sufficient to trigger robust, clocked defense activation in neighboring plants.

What is the mechanism that results in a strong activation of defenses on the onset of the second day of exposure to HIPVs? In maize, green leaf volatiles (GLVs) directly induce and prime jasmonate production and terpene release [7,8,10]. Indole can prime, but not directly induce these two responses [5]. GLVs and indole again can interact to increase defense priming upon a secondary stimulus [7]. Given these considerations, together with the observed volatile release, response kinetics, and our manipulative experiments, we can draw up the following scenario: On day one, maize plants are exposed to GLVs, which trigger a small burst of terpene release in receiver plants. At the same time, receiver plants are primed to respond more strongly on the next day, once light is restored. At the onset of the second day, four things happen simultaneously. First, the emission of GLVs from the neighboring plants increases. Second, the

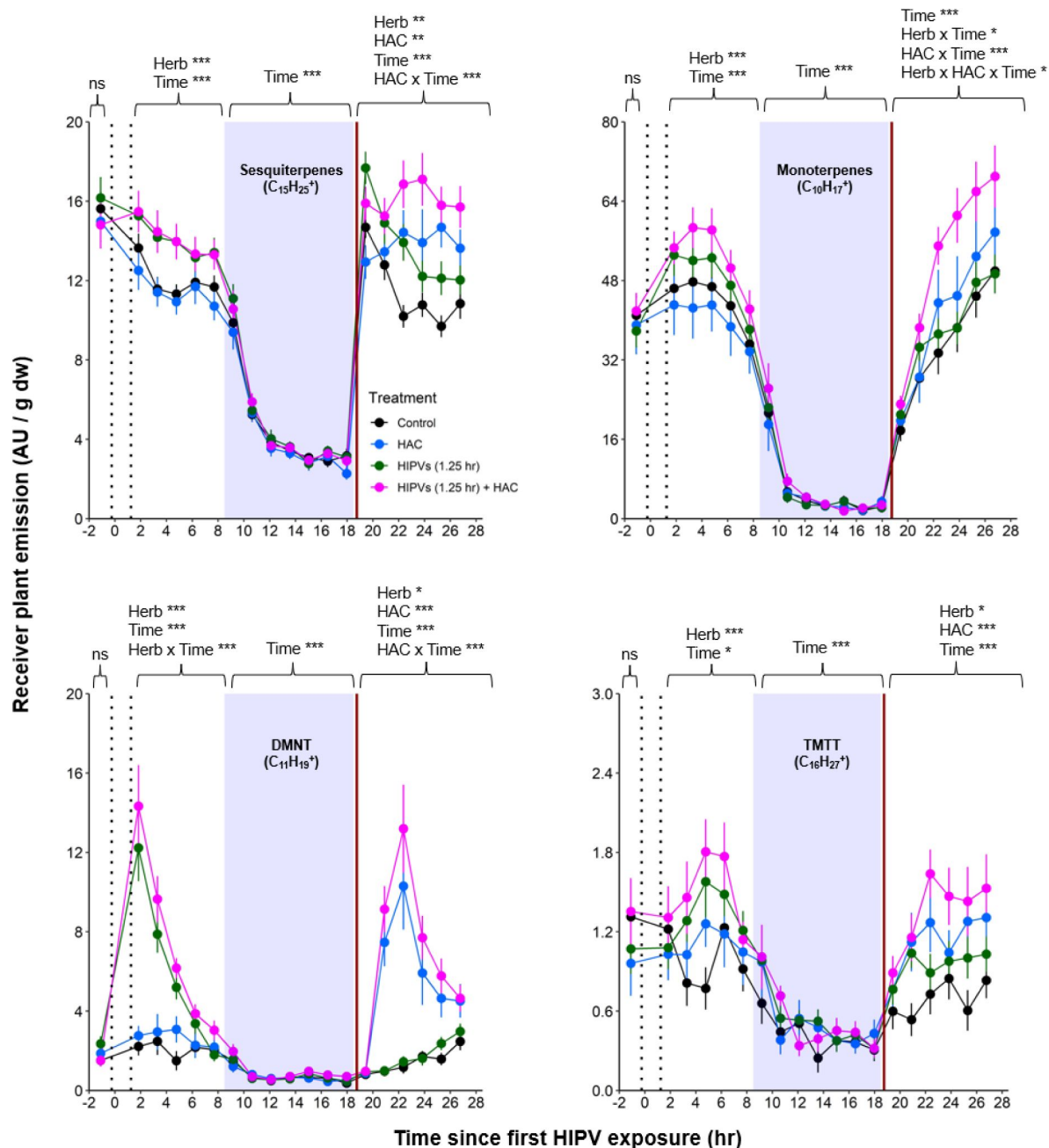


Figure 7.

The combination of volatile priming and sender emission kinetics can explain the delayed terpene burst in receiver plants. Sender plants were connected to receiver plants 30 min prior to herbivore exposure on sender plants and left connected for 1.25 hr following exposure (time between perforated vertical lines). After 1.25 hr, chambers were disconnected and measurements were collected from receiver plant chambers only. The following day, after light was restored, plants were treated with (Z)-3-hexenyl acetate (HAC) dispensers to simulate bioactive signals (indicated by red solid vertical line). * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$ as determined by aligned rank transformed nonparametric factorial repeated measures ANOVA. Abbreviations: HIPV, herbivore-induced plant volatile; DMNT, 4,8-dimethylnona-1,3,7-triene; TMTT, 4,8,12-trimethyltrideca-1,3,7,11-tetraene. Colored points represent mean emissions standardized by fresh weight (fw). Error bars = SE. $n = 12-16$. Compounds were identified based on their molecular weight + 1, as all compounds were protonated. Sesquiterpenes: $m/z = 205.20$, Monoterpenes: $m/z = 137.13$, DMNT: $m/z = 151.15$, TMTT: $m/z = 219.21$.

	Herb			HAC			Time			Herb x HAC			Herb x Time			Hac x Time			Herb x HAC x Time		
	<i>F</i>	<i>p</i>	df	<i>F</i>	<i>p</i>	df	<i>F</i>	<i>p</i>	df	<i>F</i>	<i>p</i>	df	<i>F</i>	<i>p</i>	df	<i>F</i>	<i>p</i>	df	<i>F</i>	<i>p</i>	df
SQT																					
Base	0.22	0.64	1,54	0.23	0.62	1,54	NA	NA	NA	0.05	0.82	1,54	NA	NA	NA	NA	NA	NA	NA	NA	NA
D1	13.8	<0.001	1,54	0.40	0.53	1,54	8.03	<0.001	4,216	0.30	0.59	1,54	1.85	0.12	4,216	0.10	0.98	4,216	0.23	0.92	4,216
Dark	2.15	0.15	1,54	0.17	0.69	1,54	118	<0.001	6,324	0.28	0.60	1,54	1.65	0.13	6,324	1.06	0.39	6,324	0.83	0.55	6,324
D2	8.39	0.005	1,54	8.69	0.005	1,54	12.8	<0.001	5,270	0.002	0.97	1,54	1.22	0.30	5,270	23.9	<0.001	5,270	0.93	0.46	2,570
MNT																					
Base	0.05	0.82	1,54	0.01	0.94	1,54	NA	NA	NA	1.33	0.25	1,54	NA	NA	NA	NA	NA	NA	NA	NA	NA
D1	7.84	0.007	1,54	0.02	0.88	1,54	36.5	<0.001	4,216	2.03	0.16	1,54	1.17	0.32	4,216	0.47	0.76	4,216	0.83	0.51	4,216
Dark	2.75	0.10	1,54	1.41	0.24	1,54	28.1	<0.001	6,324	2.62	0.11	1,54	1.04	0.40	6,324	1.04	0.40	6,324	1.02	0.41	6,324
D2	4.82	0.03	1,54	5.73	0.02	1,54	197	<0.001	5,270	1.68	0.20	1,54	2.57	0.03	5,270	9.90	<0.001	5,270	2.82	0.02	5,270
DMNT																					
Base	0.60	0.44	1,54	0.91	0.34	1,54	NA	NA	NA	3.52	<u>0.07</u>	1,54	NA	NA	NA	NA	NA	NA	NA	NA	NA
D1	66.3	<0.001	1,54	2.55	0.12	1,54	81.4	<0.001	4,216	0.69	0.41	1,54	65.1	<0.001	4,216	0.62	0.65	4,216	1.10	0.36	4,216
Dark	2.34	0.13	1,54	0.45	0.50	1,54	10.7	<0.001	6,324	0.51	0.48	1,54	1.05	0.40	6,324	0.20	0.98	6,324	0.30	0.94	6,324
D2	7.01	0.01	1,54	155	<0.001	1,54	52.4	<0.001	5,270	3.67	<u>0.06</u>	1,54	1.74	0.13	5,270	54.6	<0.001	5,270	2.09	<u>0.07</u>	5,270
TMTT																					
Base	0.12	0.73	1,54	0	0.99	1,54	NA	NA	NA	2.23	0.14	1,54	NA	NA	NA	NA	NA	NA	NA	NA	NA
D1	8.91	0.004	1,54	0.67	0.42	1,54	3.05	0.02	4,216	0.01	0.92	1,54	1.43	0.23	4,216	0.58	0.68	4,216	0.43	0.79	4,216
Dark	2.68	0.11	1,54	0.33	0.57	1,54	7.67	<0.001	6,324	1.43	0.24	1,54	1.22	0.30	6,324	0.46	0.84	6,324	2.00	<u>0.07</u>	6,324
D2	5.83	0.02	1,54	15.2	<0.001	1,54	5.28	<0.001	5,270	0.01	0.93	1,54	0.35	0.88	5,270	1.69	0.14	5,270	1.03	0.40	5,270

Table 2.

Aligned rank transformed nonparametric factorial repeated measures ANOVA results from data presented in **Fig 7**. Bold values: $p < 0.05$ and underlined values: $p < 0.1$. Abbreviations: HAC, (Z)-3-hexenyl acetate; SQT, sesquiterpenes; MNT, monoterpenes; DMNT, 4,8-dimethylnona-1,3,7-triene ; TMTT, 4,8,12-trimethyltrideca-1,3,7,11-tetraene.

defense priming mechanism responding to these cues kicks into gear. Third, terpene production is activated. Fourth, stomata open and enable volatile diffusion. Together, these elements result in a strong terpene burst. The orchestration of these elements is noteworthy and results in a predictable response pattern under variable HIPV exposure. Further experiments will reveal whether similar patterns are observed in other plants, and how they may operate on a mechanistic level. Additionally, considering we observed an induction in jasmonates coinciding with terpene bursts. In the future, it will be interesting to explore the kinetics of non-volatile jasmonate-dependent defenses, as they are likely to follow similar patterns. Our experiments reveal the importance of tracking plant defense responses in real time as they perceive the dynamic natural volatile blend of herbivore attacked neighbors.

What is the ecological relevance of the clocked temporal kinetics of defense activation in neighboring plants? We propose several hypotheses. First, responding most strongly to volatile cues on the second day may avoid unnecessary energy expenditure [13]. GLVs are emitted upon both herbivore attack and mechanical damage [39]. However, when damage is not sustained, GLVs will dissipate rapidly [15]. Thus, responding most strongly to repeated GLV exposure could avoid false negatives and allow plants to respond more robustly to the presence of actual herbivory. Second, sending out volatiles on the second day may maximize indirect defenses. Attracting natural enemies too early to herbivore-free plants could have fitness costs for both the plant and herbivore natural enemies. Responding on the second day could be advantageous, as by this time, the natural enemies would have located and interacted with the herbivores on sender plants, and would thus be ready to move over to the next plant in anticipation of herbivore arrival. As fitness outcomes are likely highly nuanced, especially considering HIPVs also attract herbivores [40], investigation into information transfer on the multi-plant scale, namely in the context of multi-trophic interactions, will be critical. Considerable work will be needed to understand whether the observed defense activation pattern has any adaptive benefit.

Airborne information transfer has the potential to play a major role in plant communities, as it functions as a signaling viaduct between plants not physically connected to one another [4,7,41]. Although it has been established for decades that plants transmit airborne chemical information between individuals, we are only scratching the surface of the dynamic nature of this phenomenon. The kinetics of information transfer at the detailed temporal resolution of this study provide some insights regarding how HIPVs act across time and space. On this basis, a more comprehensive understanding of volatile-mediated information transfer between plants can be built.

Methods

Plant and insect growth

V2-stage *Zea mays* (maize, B73) was used throughout this study. At this stage, maize has 4 leaves: two are fully developed, one is expanding and one is emerging. Maize plants were grown in commercial potting soil (Selmaterra, BiglerSamen, Switzerland) in 180 ml pots. Plants were grown in a greenhouse supplemented with artificial lights ($ca. 300 \mu\text{mol m}^{-2} \text{s}^{-1}$). In between light and dark phases, plants were supplemented with lower light ($ca. 60 \mu\text{mol m}^{-2} \text{s}^{-1}$) for 15 min to more accurately simulate day : night transitions. The greenhouse was maintained at $22 \pm 2^\circ\text{C}$, 40–60% relative humidity, with a 14 h : 10 h, light : dark cycle.

Two-week-old maize seedlings were used for all experiments. *Spodoptera exigua* (Frontier Agricultural Sciences, USA) were reared from eggs on artificial diet [42] and used for experiments when they reached the 4th instar stage.

Herbivore treatment and experimental setup

The herbivore treatment consisted of adding three 4th instar larvae onto plants (sender plants). For controls, sender plants were left undamaged. Two principal experimental setups were used to determine the kinetics of herbivore-induced plant volatile (HIPV) emissions in both sender plants (damaged by herbivores) and receiver plants (exposed to HIPVs from sender plants):

Setup 1

Single plants (senders) in transparent glass chambers (Ø×H 12 × 45 cm) were connected with PTFE tubing to a second transparent glass chamber that was either left empty or filled with a second plant (receiver). Chambers were sealed other than an airflow inlet on the first chamber and an air outlet on the second chamber, allowing HIPVs from sender plants to pass through the second chamber. Cumulative HIPV emissions from senders and receivers were measured as described in the section “Volatile emission sampling”. In order to determine emissions from receiver plants alone, HIPV emissions from sender plants connected to an empty chamber were subtracted from sender plants connected to receiver plants (Supplemental Fig 3).

Setup 2

To isolate emissions solely from receiver plants, chambers containing sender and receiver plants were initially kept separate for basal HIPV emission profiling. Chambers were then connected as in setup 1 for 30 min before herbivore treatment. After herbivores were added to sender plants, chambers remained connected for 1.25 hr, after which time sender and receiver plants were once again disconnected and rearranged so that receiver plants had both clean air flowing through the inlet and an outlet for volatile collection and profiling (Supplemental Fig 3).

Preparation of (Z)-3-hexenyl acetate dispensers

(Z)-3-hexenyl acetate dispensers were prepared as previously described with modifications [7,10]. In brief, 1.5-ml glass vials (Ø×H 11.6 × 32 mm) containing *ca.* 100 mg glass wool were filled with 200 µl (Z)-3-hexenyl acetate (HAC; >98%; Sigma-Aldrich, Buchs, Switzerland) diluted 50 fold in EtOH and sealed with screw caps containing a rubber septum. The caps were pierced with a 1-µl glass capillary and sealed with PTFE tape. Vials were then covered with aluminum foil and equilibrated for at least 5 days before using. The (Z)-3 isomers of all GLVs measured in this study (hexenyl acetate, hexenal, and hexen-1-ol) are known to be bioactive [8,43] and interconvertible, with hexenal and hexenyl acetate for instance being transformed into hexen-1-ol. As such, we designed the dispensers to emit a molar concentration of HAC comparable to the molar concentration of all (Z)-3 GLVs emitted by herbivore-infested plants. HAC dispensers emitted HAC at a rate of 4.68 (±0.40 SD) nmol/hr and maize seedlings infested with 3 4th-instar *S. exigua* larvae emitted all GLVs at a rate of 1.94 nmol/hr (±0.25 SD). Of note, GLV emissions induced by caterpillars vary over time, and can be more than 2-fold higher than the average during times of strong active feeding (Supplemental Fig 4). Thus, the release rate of the dispensers is within the plant’s physiological range.

Volatile sampling

Volatile emissions were sampled using setup 1 and 2. Volatile profiling was measured with a high-throughput platform comprised of a proton transfer reaction time-of-flight mass spectrometer (PTR-ToF-MS; ToFwerk, Switzerland) and an automated headspace sampling system (Abon Life Sciences, Switzerland) supplied with clean airflow (0.8 L min⁻¹). An outlet on receiver plants was accessible to the autosampler/PTR-ToF-MS system. The PTR-ToF-MS system drew air at 0.1 L min⁻¹. Between samples, a zero gas measurement was performed for 3 s to avoid contamination. At each time point volatiles were continuously measured for 15-25 s and averaged to a single mean per sample. Complete mass spectra (0- 500 m/z) were recorded in positive mode at *ca.* 10 Hz. The PTR was operated at 100 °C and an E/N of approximately 120 Td. The volatile data extraction and

processing were conducted using Tofware software package v3.2.2 (Tofwerk, Switzerland). Protonated compounds were identified based on their molecular weight + 1. During volatile collection LED lights (DYNA, heliospectra) were placed *ca.* 80 cm above the glass cylinders and provided light at *ca.* 300 $\mu\text{mol m}^{-2} \text{s}^{-1}$. Identical light : dark cycle timing as in the greenhouse for plant growth was used. In order to quantify GLVs emitted from herbivore-infested plants and HAC dispensers, we collected total volatile emissions for 1 hr using an identical volatile collection and analysis method described in Erb et al., 2015. In brief, volatiles were collected in Super-Q adsorbent traps, which were then eluted in dichloromethane and injected into a GC-MS (Agilent, USA). Compound identities were confirmed with mass spectroscopy analysis and similarity to library matches (NIST search 2.2 Mass Spectral Library, USA). Emission rates of the Z-3 isomers of GLVS were quantified using a standard curve with synthetic compounds.

Foliar terpene pools and gene expression

Setup 1 was used to determine terpene pools and gene expression. After 3, 8, 16.75 and 22 hr of exposure to HIPVs, the oldest developing leaf of receiver plants was harvested and flash frozen on liquid nitrogen. The oldest developing leaf was chosen as it is the largest and highly responsive to bioactive HIPVs [10]. A new set of plants was used for each time point. Analysis of foliar terpene pools was conducted using slightly modified, previously described, methods [44]. In brief, *ca.* 15 mg of ground fresh frozen leaf tissue was added to a 20-ml precision thread headspace glass vial sealed with a magnetic screw cap fitted with a silicone/PTFE septum (Gertel GmbH & Co. KG, Germany). Immediately after a vial was prepared, an SPME fiber (100 μm polydimethylsiloxane coating; Supelco, USA) was inserted into the vial and volatiles were collected for 40 min at 50 °C. After collection, volatiles were thermally desorbed for 3 min at 220 °C and analyzed using GC-MS (Agilent, USA). Helium was used as the carrier gas at a flow-rate of 1 ml min⁻¹ with a temperature gradient of 5 °C/min from 60 °C (1 min hold) to 250 °C. Compound identification was based on similarity to library matches (NIST search 2.2 Mass Spectral Library, USA). For quantification of gene expression, total RNA was extracted and purified from *ca.* 80 mg ground fresh frozen tissue using the GeneJET plant RNA extraction kit following the manufacturer's instructions.

Genomic DNA was removed from 1 μg purified RNA using gDNA Eraser (PrimeScript RT Reagent Kit, Perfect Real Time) following manufacturer's instructions (Takara Bio Inc., Kusatsu, Japan). Reverse transcription and cDNA was synthesized using PrimeScript Reverse Transcriptase (TaKaRa Bio). Gene expression was determined with quantitative reverse transcription polymerase chain reaction (qRT-PCR) using ORA SEE qPCR Mix (highQu GmbH, Germany) on an Applied Biosystems QuantStudio 5 Real-Time PCR system. The normalized expression (NE) values were calculated as: $NE = (1/(PE^{Ct_{\text{target}}})) / (1/(PE^{Ct_{\text{reference}}}))$ where PE = primer efficiency and Ct = cycle threshold [45]. Ubiquitin (UBI1) was used as the reference gene. Gene identifiers and primer sequences are listed in Supplemental Table 1.

Phytohormone quantification

Plant treatments and tissue collection for phytohormone were identical to foliar terpene, and gene expression analysis. The phytohormones jasmonic acid (JA), JA-isoleucine (JA-Ile) and 12-oxo-phytodienoic acid (OPDA) were extracted, analyzed and quantified using a slightly modified version of the method detailed in [46]. In brief, *ca.* 80 mg of finely ground fresh frozen leaf tissue was extracted with 1 ml ethylacetate:formic acid (99.5:0.5, v/v) spiked with isotopically labelled forms of the abovementioned phytohormones. d5-JA was acquired from CDN Isotopes (Canada), d5-OPDA was acquired from OLChemIm (Czechia), and ¹³C6-JA-Ile was synthesized at the University of Neuchâtel according to a previously described method [47]. Extracts were dried and re-suspended in 200 μl 50% MeOH using a sonicating bath for 30 min. Phytohormones were analyzed using UPLC-MS-MS fitted with an Acuity BEH C18 column (Waters, USA) using a flow-rate of 400 $\mu\text{l}/\text{min}$ and an injection volume of 2 μl . The two mobile phases used were 0.05%

formic acid in water (A) and 0.05% formic acid in acetonitrile (B) with the following gradient conditions: 5-50% B over 5 min, 60-100% B over 3 min, 100% B for 4 min and a final re-equilibration at 5% B for 4 min.

Statistical analyses

All statistical analyses were performed in R version 4.2.2 [48]. Volatile emissions were analyzed by aligned rank transformed nonparametric factorial repeated measures ANOVA using the package ARTool, as individual plant emission kinetics were measured repeatedly over time [49]. Internal terpene pools, gene expression and phytohormone levels were analyzed using Welch's *t*-tests between control and HIPV-exposed receiver plants within each time point.

Data availability statement

All data used to generate figures and tables presented in this study are available as supplemental material (Appendix A).

Acknowledgements

We would like to thank Dr. Bernardus Schimmel for assistance with gene expression quantification. Additionally, we thank all members of the Biotic Interactions and Chemical Ecology groups at the University of Bern for helpful discussions. This work was supported by the Swiss National Science Foundation (Grants Nr. 210651 and 200355), the State Secretariat for Education, Research, and Innovation SERI (Project CANWAS), the Horizon 2020 Marie Skłodowska-Curie Actions (Grant Nr. 886651) and the University of Bern.

Author Contributions

JMW and ME conceived the project. JMW, TC, LW and ME designed experiments. JMW and TMC conducted experiments. JMW, GG and ME analyzed and interpreted data. JMW wrote the initial draft of the manuscript with critical revisions from ME and input from all authors.

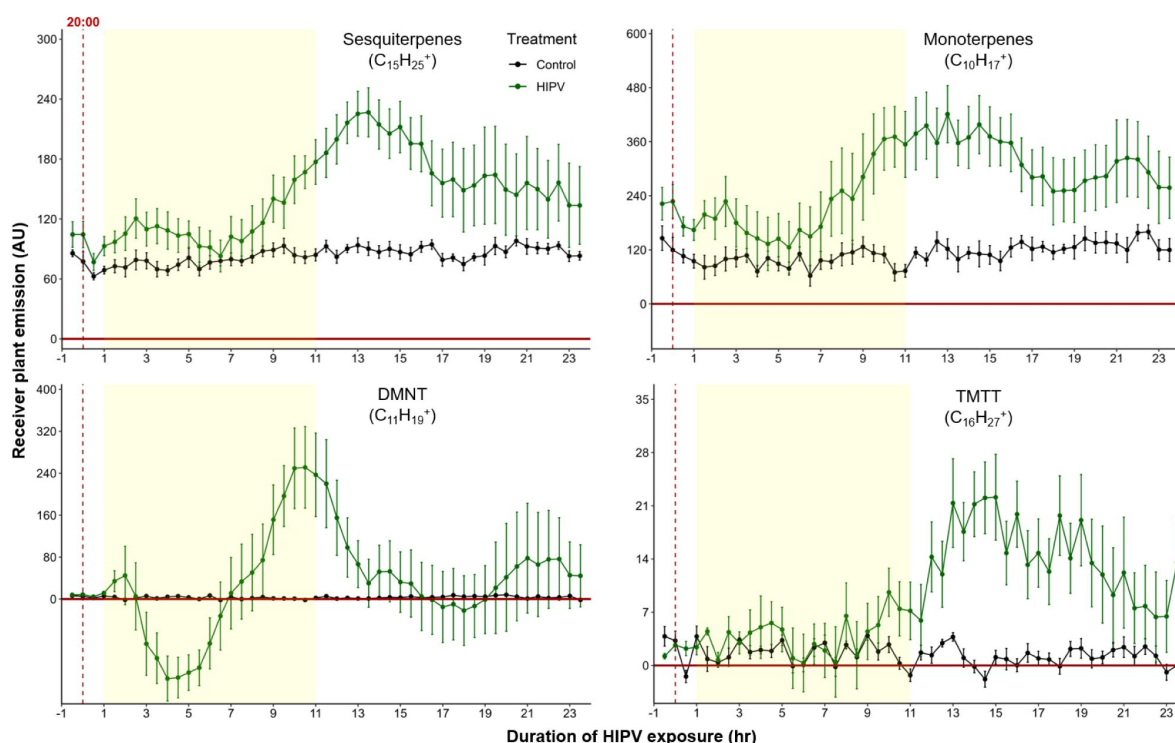
Supplemental Tables

Supplemental Figures

Target gene	Gene name	Identifier	fw	rv
TPS2	Terpene synthase 2	Zm00001d015053	TACCGGGTCGAGATCACCAA	TCGTTCTGTAACGGTGTGGAG
TPS10	Terpene synthase 10	Zm00001d024486	TGACAGCCTTGATCACCGTA	AGTTCATCCCCACACGAATC
FPPS3	Farnesyl pyrophosphate synthase 3	Zm00001d043727	CCTGGCTAGTTGTGCAAGCT	CAAAACAGTTTGGACTGCCT
CYP92C5	Dimethylnonatriene/trimethyltridecatetraene synthase	Zm00001d018839	AGGGGTTCGAAGCGGAAGATG	AGGTCAGTCGCCACAAACTC
OPR7	Oxo-phytodienoate reductase 7	Zm00001d032049	GCACGCATGTGATTGATTATTAGT	CAA TCG CGG CAT TAC CCA GAT GT
UBI1	Ubiquitin	Zm00001d015327	TAAGCTGCCGATGTGCCTGCG	CTGAAAGACAGAACATAATGAGCACAG

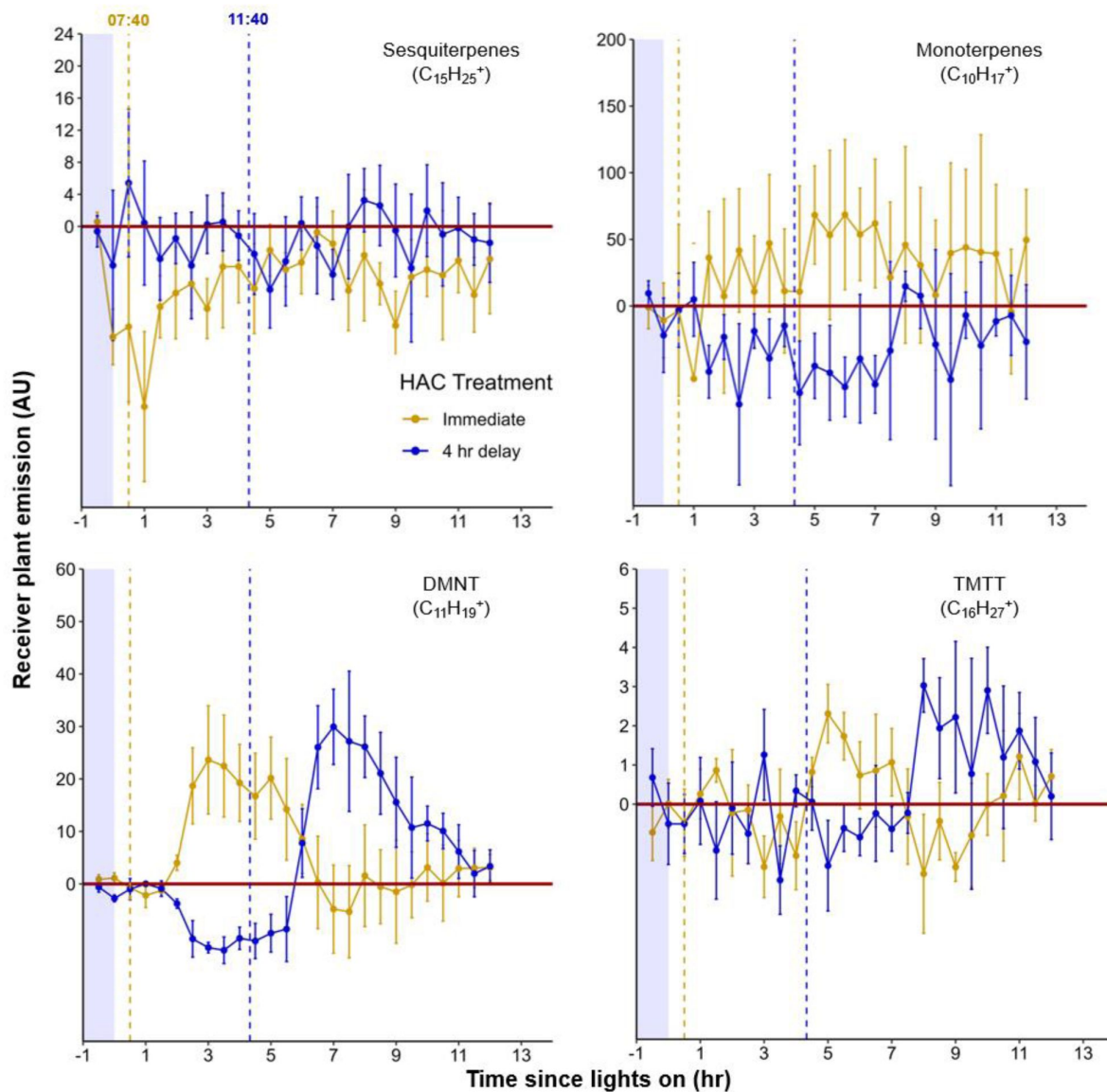
Supplemental Table 1.

Gene identifiers and qRT-PCR primer sequences used in this study.



Supplemental Figure 1.

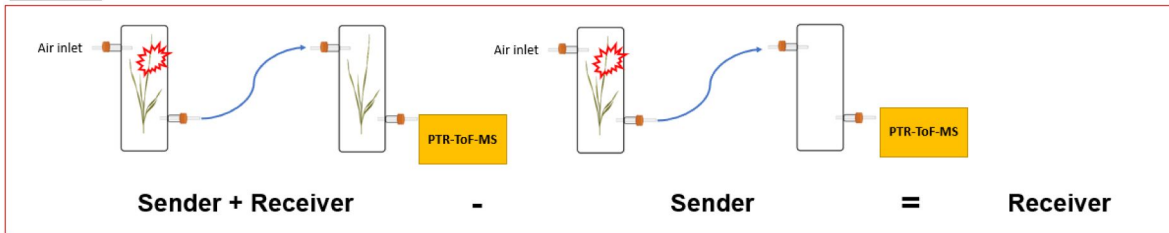
The delayed volatile burst is conserved under continuous light, regardless of the time of day. Emission kinetics of herbivore-induced plant volatile (HIPV)-induced terpenes in undamaged receiver plants under continuous light are shown. Herbivory on sender plants *ca.* 1-hr before lights would typically be turned off (around 20:00 hr). Plants were grown under normal light conditions. Lights were left on continuously following the start of the treatment. Dark green points represent the mean emission of herbivore damaged sender plants connected to undamaged receiver plants with the emissions from damaged sender plants only subtracted. Black points represent undamaged sender plants connected to undamaged receiver plants, with the emissions from undamaged sender plants only subtracted. Yellow rectangles represent when the lights would typically be turned off. Abbreviations: DMNT, 4,8- dimethylnona-1,3,7-triene; TMTT, 4,8,12-trimethyltrideca-1,3,7,11- tetraene. Error bars = SE. $n = 6$. Compounds were identified based on their molecular weight + 1, as all compounds were protonated. Sesquiterpenes: $m/z = 205.20$, Monoterpenes: $m/z = 137.13$, DMNT: $m/z = 151.15$, TMTT: $m/z = 219.21$.



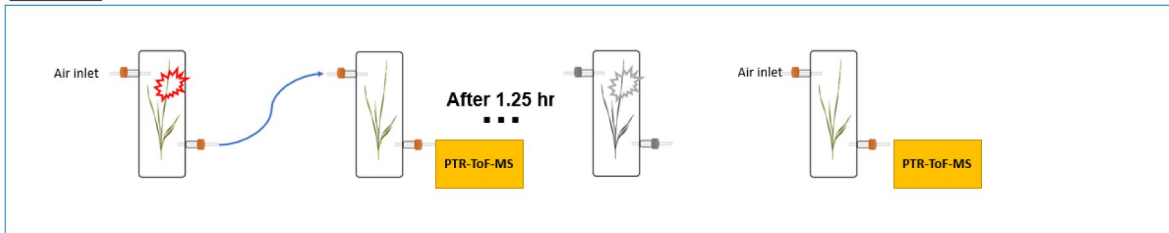
Supplemental Figure 2.

Maize seedlings respond consistently to green leaf volatiles (GLVs) throughout the day. Seedlings were exposed to (Z)-3-hexenyl acetate (HAC) either at 07:40 hr, *ca.* 30 min after lights came on (Immediate, yellow points) or at 11:40 hr, *ca.* 4 hr and 30 min after (4-hr delay, blue points). Control (untreated) emission levels were subtracted from both treatments. Yellow and blue perforated vertical lines represent the moment that HAC dispensers were added to the immediate and 4-hr delay plants, respectively. Abbreviations: DMNT, 4,8- dimethylnona-1,3,7-triene; TMTT, 4,8,12-trimethyltrideca-1,3,7,11- tetraene. Error bars = SE. $n = 4$. Compounds were identified based on their molecular weight + 1, as all compounds were protonated. Sesquiterpenes: $m/z = 205.20$, Monoterpenes: $m/z = 137.13$, DMNT: $m/z = 151.15$, TMTT: $m/z = 219.21$.

Setup 1:

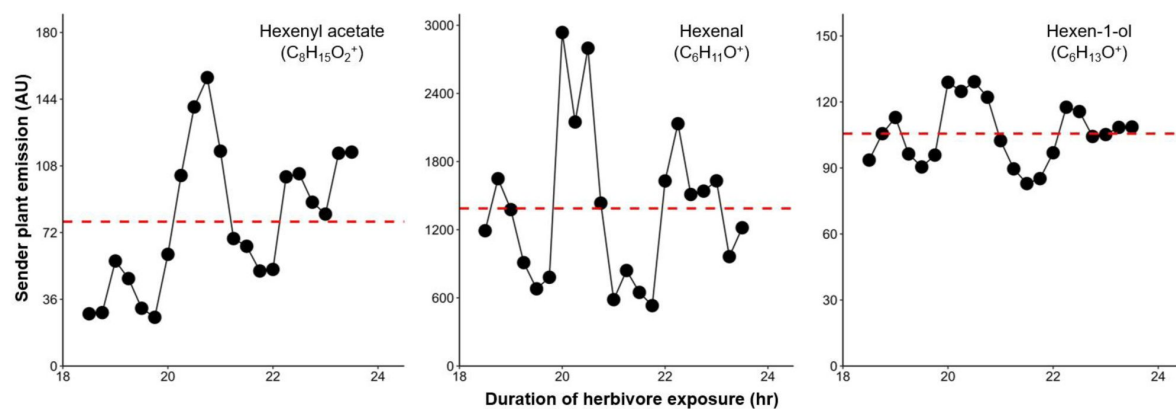


Setup 2:



Supplemental Figure 3.

Experimental setup schemes for volatile profiling



Supplemental Figure 4.

Green leaf volatile (GLV) emission from herbivore-damaged maize seedlings is highly variable over time. Representative GLV emissions from a single attacked plant on the second day of herbivory (18-24 hr of exposure to HIPVs) are shown in black. The perforated red horizontal line represents the average across all time points presented. Compounds were identified based on their molecular weight + 1, as all compounds were protonated. Sesquiterpenes: $m/z = 205.20$, Monoterpenes: $m/z = 137.13$, DMNT: $m/z = 151.15$, TMTT: $m/z = 219.21$.

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Reviewer #1 (Public Review):

The authors of the manuscript "High-resolution kinetics of herbivore-induced plant volatile transfer reveal tightly clocked responses in neighboring plants" assessed the effects of herbivory induced maize volatiles on receiver plants over a period of time in order to assess the dynamics of the responses of receiver plants. Different volatile compound classes were measured over a period of time using PTR-ToF-MS and GC-MS, under both natural light:dark conditions, and continuous light. They also measured gene expression of related genes as well as defense related phytohormones. The effects of a secondary exposure to GLVs on primed receiver plants was also measured.

The paper addresses some interesting points, however some questions arise regarding some of the methods employed. Firstly, I am wondering why VOCs (as measured by GC-MS) were not quantified. While I understand that quantification is time consuming and requires more work, it allows for comparisons to be made between lines of the same species, as well as across other literature on the subject. Simply relying on the area under the curve and presenting results using arbitrary units is not enough for analyses like these. AU values do not allow for conclusions regarding total quantities, and while I understand that this is not the main focus of this paper, it raises a lot of uncertainty for readers (for example, the references cited show that TMTT has been found to accumulate at similar levels of caryophyllene, however the AU values reported are an order of magnitude higher for TMTT. Again, without actual quantification this is meaningless, but for readers it is confusing).

With regards to the correlation analyses shown in figure 6, the results presented in many of the correlation plots are not actually informative. While there is a trend, I do not think that this is an appropriate way to show the data, as there are clearly other relationships at play. The comparison between plants under continuous light and normal light:dark conditions is interesting.

This paper addresses a very interesting idea and I look forward to seeing further work that builds on these ideas.

Reviewer #2 (Public Review):

The exact dynamics of responses to volatiles from herbivore-attacked neighbouring plants have been little studied so far. Also, we still lack evidence whether herbivore-induced plant volatiles (HIPVs) induce or prime plant defences of neighbours. The authors investigated the volatile emission patterns of receiver plants that respond to the volatile emission of neighbouring sender plants which are fed upon by herbivorous caterpillars. They applied a very elegant approach (more rigorous than the current state-of-the-art) to monitor temporal response patterns of neighbouring plants to HIPVs by measuring volatile emissions of senders and receivers, senders only and receivers only. Different terpenoids were produced within 2 h of such exposure in receiver plants, but not during the dark phase. Once the light turned on again, large amounts of terpenoids were released from the receiver plants. This may indicate a delayed terpene burst, but terpenoids may also be induced by the sudden change in light. As one contrasting control, the authors also studied the time-delay in volatile emission when plants were just kept under continuous light. Here they also found a delayed terpenoid production, but this seemed to be lower compared to the plants exposed to the day-night-cycle. Another helpful control was now performed for the revision in which the herbivory treatment was started in the evening hours and lights were left on. This experiment revealed that the burst of terpenoid emission indeed shifted somewhat. Circadian and diurnal processes must thus interact.

Interestingly, internal terpene pools of one of the leaves tested here remained more comparable between night and day, indicating that their pools stay higher in plants exposed to HIPVs. In contrast, terpene synthases were only induced during the light-phase, not in the dark-phase. Moreover, jasmonates were only significantly induced 22 h after onset of the volatile exposure and thus parallel with the burst of terpene release.

An additional experiment exposing plants to the green leaf volatile (glv) (Z)-3-hexenyl acetate revealed that plants can be primed by this glv, leading to a stronger terpene burst. The results are discussed with nice logic and considering potential ecological consequences. All data are now well discussed.

Overall, this study provides intriguing insights in the potential interplay between priming and induction, which may co-occur, enhancing (indirect and direct) plant defence. Follow-up studies are suggested that may provide additional evidence.

<https://doi.org/10.7554/eLife.89855.2.sa0>

Author Response

The following is the authors' response to the current reviews.

Reviewer #1 (Public Review):

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This paper addresses a very interesting idea and I look forward to seeing further work that builds on these ideas.

As mentioned in our previous response, we have added the quantification of GLVs in order to increase the comparability of our work to other studies.

Regarding the comment about TMTT (only measured as internal pools), the purpose of the inclusion of these internal pool data, was simply to determine whether terpenes were accumulating in leaf tissue during the night when emissions are hindered (likely due to closed stomata). The data clearly show that internal terpene pools do not accumulate above daytime levels during darkness – this is further supported by gene expression data that show downregulation of terpene synthase genes during darkness. While quantification would certainly increase the ability to compare internal pools, it would not change the interpretation of our results. Also note that absolute quantification is challenging for compounds such as TMTT, which are not readily available.

Regarding the comment on Figure 6, while we agree there may be interesting patterns beyond linear relationships, as stated in our previous response, the purpose of our analysis was to determine if the higher terpene burst in receiver plants on the second day may be explained by sender plants emitting more GLVs on the second day. Figure 6 shows that this is not the case. Further analyses would not provide additional significant insights into the hypothesis that we tested here.

We thank the reviewer for their overall positive outlook on our paper and for the constructive comments.

Reviewer #2 (Public Review):

The exact dynamics of responses to volatiles from herbivore-attacked neighbouring plants have been little studied so far. Also, we still lack evidence whether herbivore-induced plant volatiles (HIPVs) induce or prime plant defences of neighbours. The authors investigated the volatile emission patterns of receiver plants that respond to the volatile emission of neighbouring sender plants which are fed upon by herbivorous caterpillars. They applied a very elegant approach (more rigorous than the current state-of-the-art) to monitor temporal response patterns of neighbouring plants to HIPVs by measuring volatile emissions of senders and receivers, senders only and receivers only.

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Overall, this study provides intriguing insights in the potential interplay between priming and induction, which may co-occur, enhancing (indirect and direct) plant defence. Follow-up studies are suggested that may provide additional evidence.

We thank the reviewer for their positive outlook on our paper and for their constructive comments.

Recommendations for the authors:

Reviewer #2 (Recommendations For The Authors):

The authors did a great job with the revision. The additional experiments strengthened their conclusions. Thanks also for performing the suggested test for potential differences in induction capacity at different times of day, the new data are very interesting.

Thank you very much.

Line 49-52: The newly added sentence could be clarified in wording.

We will clarify the sentence.

Line 254-255: The newly added sentence needs to be corrected. This is no full sentence and it is not clear what the authors wanted to say here.

We will clarify this sentence.

Figure 6: In those instances, in which the correlation is not significant, the line should not be shown.

We will remove the lines when correlations are not significant.

The names of chemical compounds and terpene synthases should be written in lower case letters (see legend Fig 6, e.g. hexenal, not Hexenal; legend fig. 2: terpene synthase,

not Terpene synthase)

In the last round of revisions, I commented on Line 23: consequences on community dynamics are not investigated here, so this is a bit misleading. ... Your response was "We have deleted the sentence about community dynamics ..." which, however, in fact was not done! Please change!

Apologies for that, we will delete mention of community dynamics in that sentence (for real).

The following is the authors' response to the original reviews.

eLife assessment

This important study examines the effects of herbivory-induced maize volatiles on neighboring plants and their responses over time. Measurements of volatile compound classes and gene expression in receiver plants exposed to these volatiles led to the conclusion that the delayed emission of certain terpenes in receiver plants after the onset of light may be a result of stress memory, highlighting the role of priming and induction in plant defenses triggered by herbivore-induced plant volatiles (HIPVs). Most experimental data are compelling but additional experiments and accurate quantifications of the compounds would be required to confirm some of the main claims.

Response: We thank the editors for their overall positive feedback on our MS. We have added additional experiments to quantify green leaf volatile emissions in both sender plants and synthetic dispensers (Reviewer 1) and address the importance of the precise time of day plants are induced (Reviewer 2). These additions strengthen the main conclusions of our study.

Public Reviews:

Reviewer #1 (Public Review):

The authors of the manuscript "High-resolution kinetics of herbivore-induced plant volatile transfer reveal tightly clocked responses in neighboring plants" assessed the effects of herbivory-induced maize volatiles on receiver plants over a period of time in order to assess the dynamics of the responses of receiver plants. Different volatile compound classes were measured over a period of time using PTR-ToF-MS and GC-MS, under both natural light:dark conditions, and continuous light. They also measured gene expression of related genes as well as defence-related phytohormones. The effects of a secondary exposure to GLVs on primed receiver plants were also measured.

The paper addresses some interesting points, however, some questions arise regarding some of the methods employed. Firstly, I am wondering why VOCs (as measured by GC-MS) were not quantified. While I understand that quantification is time-consuming and requires more work, it allows for comparisons to be made between lines of the same species, as well as across other literature on the subject. As experiments with VOC dispensers were also used in this experiment, I find it even more baffling that the authors didn't confirm the concentration of the emission from the plants they used to make sure they matched. The references cited justifying the concentration used (saying it was within the range of GLVs emitted by their plants) to prepare the dispenser were for either a different variety of maize (delprim versus B73) or arabidopsis. Simply relying on the area under the curve and presenting results using arbitrary units is not enough for analyses like these.

Response: We thank the reviewer for their comment. We have now quantified both the emission of dispensers and maize seedlings infested with 3 4th-instar *Spodoptera exigua* larvae. Averaged across 1 h, HAC dispensers emitted roughly 2x higher molar concentrations than total GLV molar concentrations emitted by plants infested by 3 caterpillars. Of note, GLV emissions induced by caterpillars vary over time, and can be more than 2-fold higher than the average during times of strong active feeding (Supplemental Fig 4). Thus, the release rate of the dispensers is well within the plant's physiological range.

Note that the references cited were included to support the claim of the biological activity of all three GLVs rather than to justify concentration of our dispensers. We have rephrased this sentence to reflect this (see L330-333).

With regards to the correlation analyses shown in Figure 6, the results presented in many of the correlation plots are not actually informative. By blindly reporting the correlation coefficient important trends are being ignored, as there are clearly either bimodal relationships (e.g. upper left panel, HAC/TMTT, HAC/MNT) or even stranger relationships (e.g. upper left panel, IND/SQT, IND/MNT) that are not being well explained by a correlation plot. It is not appropriate to discuss the correlation factors presented here and to draw such strong conclusions on emission kinetics. The comparison between plants under continuous light and normal light:dark conditions is interesting, but I think there are better ways to examine these relationships, for example, multivariate analysis might reveal some patterns.

Response: We thank the reviewer for their comment. With our analysis we aimed at testing specifically whether the high release of known bioactive volatiles (GLVs and indole) by sender plants on the second day can explain the higher terpene emissions in the receiver plants. We explicitly mention this in the text (L176-L186). Indeed, under normal light conditions (light and dark phase), there are clear positive correlations between the GLV release of sender plants and the terpene release of receiver plants over time (see also Fig 1 and Fig 5). However, under continuous light conditions, GLV emissions in sender plants no longer correlate with terpene emissions in receiver plants (also apparent by comparison of Fig 4 and Fig 5). This shows that temporal variation in GLV emissions are insufficient to explain the delayed terpene burst. This is the relevant conclusion we draw from this analysis. As presented, we find the data to provide strong evidence that the delayed burst in receiver plant terpene emissions cannot be solely explained by higher availability of active signals on the second day. The priming experiment in Figure 7 then provides a direct additional test for this concept. While more complex analyses could indeed reveal additional patterns, these would not be particularly informative for the question at hand.

In Figure 2, the elevated concentrations of beta-caryophyllene found in the control plants at 8h and 16.75h measurement timepoints are curious. Is this something that is commonly seen in B73?

Response: We thank the reviewer for this comment. A small number of untreated plants indeed accumulated β -caryophyllene at night, which is likely the result of biological variability between samples. Our plants were soil-grown, and it is for instance possible that variation in soil biota may account for this variability. Alternatively, some plants may have been slightly stressed during handling. Note that this variability does not affect any of the conclusions in our manuscript.

While there can be discrepancies between emissions and compounds actually present within leaf tissue, it is a little bit odd that such high levels of β -caryophyllene were found at these timepoints, however, this is not reflected in the PTR-ToF-MS measurements of sesquiterpenes. It would be beneficial to include an overview of the mechanism of

production and storage of sesquiterpenes in maize leaves, which would clarify why high amounts were found only in the GC-MS analysis and not the PTR-ToF-MS analysis, which is a more sensitive analytical tool. It is possible that the amounts of β -caryophyllene present in the leaf are actually extremely low, however as the values are not given as a concentration but rather arbitrary units, it is not possible to tell. I would include a line explaining what is seen with β -caryophyllene.

Response: Thank you for this comment. It is important to note that accumulation in maize leaves can differ substantially from emission, especially at night when stomata are closed. This has been observed before in maize leaves (Seidl-Adams et al., 2015). As the reviewer suspects, earlier work indeed found that β -caryophyllene is a minor sesquiterpene compared to β -farnesene and α -bergamotene in B73 (Block et al., 2018). The PTR-ToF-MS does not discriminate between terpenes with the same m/z and thus measures total sesquiterpene emissions. Given that sesquiterpene emissions are strongly regulated by stomatal aperture and that overall sesquiterpene accumulation in control plants is low, it is not surprising that we measure only minor amounts of sesquiterpene emissions in general, and in control plants in particular. We now text to the manuscript to explain these aspects (L116-L122). Block, A.K., Hunter, C.T., Rering, C. et al. Contrasting insect attraction and herbivore-induced plant volatile production in maize. *Planta* 248, 105–116 (2018).

Seidl-Adams I, Richter A, Boomer KB, Yoshinaga N, Degenhardt J, Tumlinson JH. Emission of herbivore elicitor-induced sesquiterpenes is regulated by stomatal aperture in maize (*Zea mays*) seedlings. *Plant Cell Environ.* 38, 23-34 (2015).

Additionally, it seems like the amounts of TMTT within the leaf are extraordinarily high (judging only by the au values given for scale), far higher than one would expect from maize.

Response: We are unsure about the reviewer's interpretation here. The AU values do not allow for conclusions regarding total quantities. An earlier study found that TMTT in induced B73 plants accumulates to similar amounts as β -caryophyllene (Block et al., 2018), thus it is not surprising to detect significant TMTT pools in induced maize leaves. It is important to note that the aim of the experiment here was to test the hypothesis that plants may be hyperaccumulating volatiles when the stomata are closed at night, which could potentially explain the delayed terpene burst on the second day. We do not observe such a hyperaccumulation, thus ruling out this as the primary factor responsible for the observed phenomenon. This is further supported by the continuous light experiments, where the delayed burst in terpene emission is not hindered by the lack of a dark phase.

Block, A.K., Hunter, C.T., Rering, C. et al. Contrasting insect attraction and herbivore-induced plant volatile production in maize. *Planta* 248, 105–116 (2018).

Reviewer #2 (Public Review):

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terpenoids may also be induced by the sudden change in light. A potential caveat exists with respect to the exact timing and the day-night cycle. The timing may be critical, i.e. at which time-point after onset of light herbivores were placed on the plants and how long the terpene emission lasted before the light was turned off. If the rhythm or a potential internal clock matters, then this information should also be highly relevant. Moreover, light on/off is a rather arbitrary treatment that is practical for experiments in the laboratory but which is not a very realistic setting. Particularly with regard to terpene emission, the sudden turning on of light instead of a smooth and continuous change to lighter conditions may trigger emission responses that are not found in nature.

Response: We thank the reviewer for their comment. Although not explicitly mentioned it in the initial draft of the MS, we employed 15 min transition periods for light and dark phase transitions with a light intensity of 60 $\mu\text{mol m}^{-2} \text{s}^{-1}$ (compared to 300 $\mu\text{mol m}^{-2} \text{s}^{-1}$ at full light) to achieve a more gradual transition. We now included this information in the manuscript (L291-L292).

As one contrasting control, the authors also studied the time-delay in volatile emission when plants were just kept under continuous light (just for the experiment or continuously?). Here they also found a delayed terpenoid production, but this seemed to be lower compared to the plants exposed to the day-night-cycle. Another helpful control would be to start the herbivory treatment in the evening hours and leave the light on. If then again plants only release volatiles after a 17 h delay, the response is indeed independent of the diurnal clock of the plant.

Response: This is a very interesting point raised by the reviewer. We now conducted an additional experiment under continuous light where we started the herbivory treatment just before the start of the dark phase (ca. 20:00 PM). We found a similar pattern: a distinct delay in the highest burst. However, interestingly, the burst was shifted from 12-18 hr to 10-12 hr (Supplemental Fig 1). This burst aligned reasonably well with the point at which lights would normally be turned on again. In light of this, and, as the herbivore additions typically started ca. 5 hrs after the onset of light following a dark period (Figures 1-7), we wanted to rule out the possibility that the lack of a burst on the first day, was simply due to a difference in induction capacity depending on how shortly after the onset of light plants became exposed to GLVs. As such, we designed an additional experiment to examine whether exposure to GLVs immediately after the lights come on induce higher terpene emissions than plants exposed to GLVs ca. 5 hr after lights come on (Supplemental Fig 2). Interestingly, emissions across the terpenes were similar, regardless how long after the onset of lights on plants were exposed to GLVs. This suggests that the delayed burst is not due to the fact that, on the second day, plants are exposed to GLVs immediately after the lights come on whereas the first day they are only exposed 5 hr after the lights come on. Both continuous light experiments (normal timing and shifted timing) show bursts that occur slightly earlier than we observe with under normal day : night light conditions (L159-L166 and L207-L211), suggesting an interaction between circadian and diurnal processes. For instance, it is possible that plants would start producing volatiles slightly earlier than the onset of the day, however, light and stomatal opening limits the exact timing of the burst under normal light:dark transitions. The additional data provide further evidence for the delayed burst as a timed response in maize plants.

Additionally, we have added explanation the continuous light figure legends that plants were grown under normal conditions and lights were only left on following treatment.

Interestingly, internal terpene pools of one of the leaves tested here remained more comparable between night and day, indicating that their pools stay higher in plants exposed to HIPVs. In contrast, terpene synthases were only induced during the light-

phase, not in the dark-phase. Moreover, jasmonates were only significantly induced 22 h after the onset of the volatile exposure and thus parallel with the burst of terpene release. An additional experiment exposing plants to the green leaf volatile (glv) (Z)-3-hexenyl acetate revealed that plants can be primed by this glv, leading to a stronger terpene burst. The results are discussed with nice logic and considering potential ecological consequences. Some data are not discussed, e.g. the jasmonate and gene induction pattern.

Response: Thanks for this comment. We have added a sentence regarding the jasmonate data suggesting that, in addition to providing an additional layer of evidence for the observed delay, suggest that other JA-dependent defenses in maize may follow similar temporal patterns (L254-L257).

Overall, this study provides intriguing insights into the potential interplay between priming and induction, which may co-occur, enhancing (indirect and direct) plant defence. Follow-up studies are suggested that may provide additional evidence.

Reviewer #1 (Recommendations For The Authors):

Could the authors please explain why they chose not to calculate concentrations for VOCs? Perhaps it is that B73 is a very unique variety in that it contains very high levels of TMTT, even in control plants? This should be clarified by the authors.

Response: We address this comment in the public review portion

For the legend within Figure 2, I would move it to be in the upper left or right corners of the figure. It is not easy to see in its current position.

Response: We have moved the figure legend based on the reviewers recommendation

Figures depicting PTR-ToF-MS data: add m/z values to either the figures themselves and/or the legends.

Response: We have added m/z values to the legends and added molecular formulas of protonated compounds to each panel.

Overall, here are some other suggestions: I am slightly weary of the term "clocked response". I'm not sure this is the correct fit for what you are trying to convey. I think "regulated" is a better term than "clocked". I understand that it is likely a stylistic choice to use this word, however, I advise reconsidering for the sake of clarity of the results.

Response: Thank you. We find clocked to be an appropriate term, as it highlights the temporal aspect of the burst, and have thus left the title as is.

Have another look at the references as some are not in the correct format (i.e., species not in italics).

Response: We have checked and corrected the references

Reviewer #2 (Recommendations For The Authors):

Line 23: consequences on community dynamics are not investigated here, so this is a bit misleading.

Last sentence of the abstract: It would be nice to read the answer to this long-standing question here.

Response: We have deleted the sentence about community dynamics and provided a more concrete final sentence (L38-L40)

Lines 48-50: The example does not fit so well with the first sentence and is not entirely clear (relation to temporal dynamics; similar to what?).

Response: We have reworded the sentence for clarity (L49-L52)

Line 56: "volatiles" should be plural.

Response: Changed (L58)

Line 58: "to be produced" rather than "to produce"

Response: This seems a stylistic choice, and have left it as is.

End of abstract: Did you have any hypotheses? These should be stated here.

Response: The listing of hypotheses is also a stylistic choice, which is in some cases required by journals, but not eLife. As such we have not included a discrete list of hypotheses and instead describe what we aimed to investigate and what we found.

Line 93: "This response disappeared at night." Does this mean: "No volatiles were emitted during night"? Or was this a gradual disappearance? How many hours after the onset of light did the herbivore treatment start and how many hours after the first emission of volatiles was the light turned off?

Response: We have added when herbivory began (L92-L93) and changed the text to 'as soon as light was restored' (L97-L98).

Line 93: "as soon as the night was over" means practically rather "as soon as the light was switched on".

Response: See above

Line 91: "small induction" - do you mean "low amounts of xxx"?

Response: We mean a small induction. Terpene emission is relatively low (hence small), but still induced relative controls.

Line 91: which mono- and sesquiterpenes were monitored?

Response: It is PTR-ToF-MS and thus we cannot identify individual sesquiterpenes and monoterpenes (as they all have the same mass), and thus group them generally.

Figure 1: What exactly is the "control"? And what does the vertical hatched line in the beginning represent?

Response: We have defined the control and added a sentence describing the vertical hatched line

"Black points represent the same but with undamaged sender plants" - what is "the same" here? I find that a bit confusing!

Response: We have rephrased

Line 104: how do you define an "overaccumulation"?

Response: We have added 'above daytime levels' to clarify that we mean over daytime levels (L106)

Why was the oldest developing leaf chosen? Is this the largest one when plants are two weeks old? How many leaves do they have then? Is this the leaf with the highest biomass?

Response: We chose this leaf as it is the largest and also highly responsive to HIPVs. We have added this sentence (with a reference) in the methods section (L369-L370)

Line 107: "started increasing after 3 hours" - they may already have started before. The following description also sounds like the dynamics were investigated here. However, instead the authors measured samples at four distinct time-points and cannot say whether something "began" or "remained" etc. The wording should be changed to a more appropriate description, describing the differences at a given time-point.

Response: We changed the wording to 'were marginally induced after 3 hr' see L110

Line 113: What do you mean by "delete BELOW NIGHTTIME levels"?

Response: The word we used was 'deplete' to 'drop' (L116)

Line 114: "the expression of terpene synthases" add "in the receiver plants exposed to HIPVs."

Response: Added

Figure 2ff: The situation of receiver plants exposed to control plant volatiles is not explained in the method section and also not depicted in the Suppl. Fig. 1. Here, the sender plants seem to always have been induced (if the red star-like structure should resemble an induction - a legend may be helpful here).

Response: We have changed to 'connected to undamaged sender plants'. We additionally added a sentence to the methods section describing controls L300

Line 140: This treatment is not described in the methods section. Were the plants only kept under constant conditions for the 2 experimental days? Compared to the induction shown in Fig. 1, the amount of released volatiles seems less here.

Response: We have added explanation of this to the figure legends, explaining that plants were grown under normal conditions and lights were only left on following treatment

Another helpful control would be to start the herbivory treatment in the evening hours and leave the light on. If then again plants only release volatiles after a 17 h delay, the response is indeed independent of the diurnal clock of the plant.

Response: See public review comment. We have added this experiment and discuss it accordingly in the MS (L159-L166 and L207-L211)

| *Line 157: Check sentence/grammar*

Response: Checked and modified

| *Figure 5: I suggest using a different colour for volatiles released from the sender plants, not again the green also used in the other figures for the receiver plants. This would help the reader to quickly see which plants are in focus in each figure.*

Response: We have changed the color of the figures for clarity

| *Figure 6 legend: check grammar in several sentences (use of singular vs. plural)*

Response: We have made the tense uniform

| *The diurnal rhythm of jasmonates (and potentially also terpene synthases?) is not considered in the discussion.*

Response: See above, and we have added a sentence to the discussion mentioning the jasmonates (L254-L257)

| *Line 230-231: check grammar. Given the complexity, the response pattern may not be so predictable.*

Response: We do not understand this comment, but have checked the grammar throughout the manuscript.

| *Line 235: I like the discussion on potential ecological consequences.*

| *While some interpretation for each experiment is already given in the results section, not all results are discussed in the discussion section. For example, the jasmonate data are not discussed. This should be added.*

Response: See above

| *Line 266: To get an idea about the plant size: How many leaves do the plants have in that stage?*

Response: Added a sentence describing the size L287-L288

| *Line 321: change to "as in the greenhouse"*

Response: Changed

| *Line 334: How were the terpenoids identified and, in particular, quantified?*

Response: Added (L379-L380)

| *Line 354: Maybe rather change to: "Plant treatments and tissue collection for phytohormone sampling were identical as described above for terpene and gene expression analysis.*

Response: Changed

| *Line 357: add "material" or "leaf tissue" after "flash frozen"*

Response: Added

| *Line 359: What was the source of the isotopically labelled phytohormones?*

Response: Added (L400-L403)

| *Line 360: The phytohormones are "analyzed" using UPLC. The "quantification" is then done afterward. Please correct.*

Response: Corrected (L404)

| *Overall: a great approach and a wonderful idea!*

Thanks